

Building a Private Smart Home: User Motivations and Challenges

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A private smart home is not a finished state — people continuously renegotiate privacy as concerns and trade-offs change.

Background

- Smart homes promise convenience and automation, but many devices rely on cloud services and continuous data collection.
- Research shows that people often express concerns about privacy, yet still adopt technologies that collect personal data — a phenomenon known as the privacy paradox.
- Prior work has also shown that smart homes are typically assembled gradually over time, as households experiment with devices and adapt systems to everyday life.
- However, we still know relatively little about how privacy concerns translate into concrete decisions when building and maintaining a smart home.

How do privacy-conscious users set up and maintain privacy in their smart homes, and what raises privacy concerns in these systems?

Method

Participants
Twelve semi-structured interviews with privacy-conscious Home Assistant users, an open-source smart home platform focused on local control. Participants were primarily based in Europe and the United States, and all identified as men.

Pre-task
Before each interview, participants completed a short task listing privacy best practices for someone setting up a smart home, to prompt reflection on privacy-related decisions.

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Interviews
Semi-structured interviews explored participants' household context, smart home setups, and privacy-related decisions during setup and ongoing maintenance.

Analysis
Data were analyzed using reflexive thematic analysis, conducted by the first author.

Insights

Theme 1:
Privacy drives abandonment and replacement of smart home components

Privacy concerns can lead users to remove or replace devices, services, or entire ecosystems. As trust, life circumstances, or broader technology developments change, technologies that once felt acceptable may no longer feel comfortable to use.

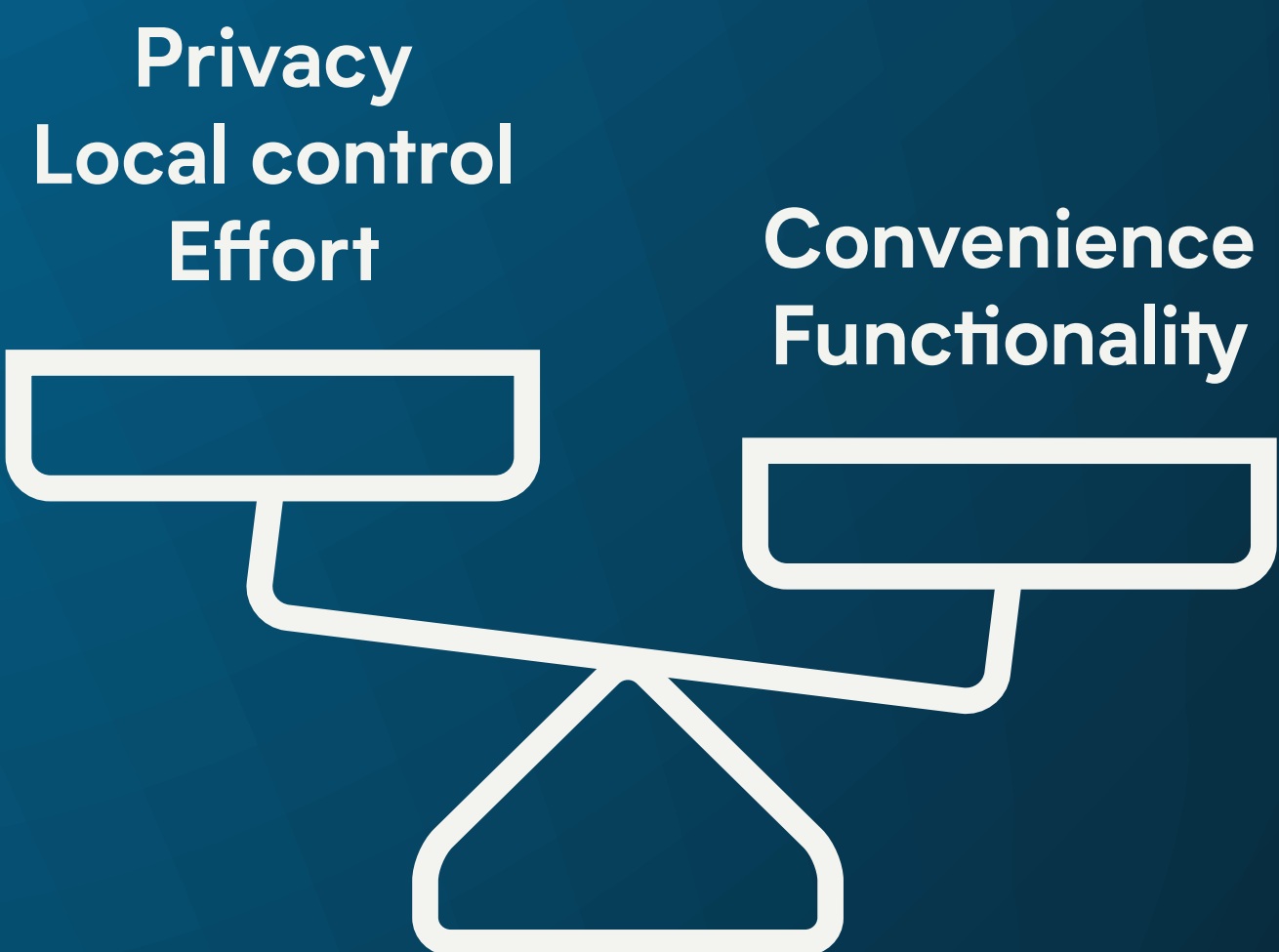


Smart home ecosystem

“Big Tech companies are more and more powerful to a point where they kind of dwarf some smaller nations.”

Theme 2:
Making the smart home private is not an end in itself

Participants aim for a functional smart home and pursue “good-enough” privacy while balancing privacy with convenience, functionality, and effort. As a result, privacy intentions and everyday technology choices do not always align.



“It’s functionality versus [...] protecting privacy as a trade off [...] at some point, I’m like: Ahh, I don’t care.”

Theme 3:
There are recurring strategies for setting up and maintaining a private smart home

Users develop strategies such as prioritizing local control, avoiding cloud services, and isolating devices on networks. However, these strategies require effort, are difficult to maintain consistently, and often leave users uncertain about how effective they are.

Design implications

Enabling privacy reassessment in changing contexts
Users reconsider privacy when circumstances or technologies change. Platforms should support users’ reflection on evolving privacy preferences.

Motivating and guiding privacy-oriented action
Platforms should provide understandable privacy assessments and concrete, prioritized actions to improve privacy.

Supporting privacy as an ongoing, visible, and actionable practice
Privacy maintenance should be supported through clearer visibility of data flows and periodic privacy checkups.

Making data flows legible through network-level feedback
Visualizing external communication can help users detect unexpected data flows and maintain privacy.

Limitations

- Participants were smart home users who are technically experienced and valued privacy, which may limit transferability to other households.
- All participants identified as men, reflecting the gender distribution of the community but limiting perspectives represented.
- Participants were primarily based in Europe and the United States, limiting cultural and regulatory diversity.